



Wholly owned by Cooperatives



TRAINING REPORT

IFFCO PARADEEP

Submitted by: SAGAR SUTAR(19CHE014)

Sutars137@gmail.com

TRAINING REPORT ON
VOCATIONAL TRAINING
Duration:15th May 22 to 30th June 22
DAP MANUFACTURING PLANT
IFFCO PARADEEP



**GANDHI INSTITUTE OF ENGINEERING
& TECHNOLOGY UNIVERSITY GUNUPUR**

Submitted to: -

R. K.PADHI

HOD Chemical Engineering

GIET UNIVERSITY,GUNUPUR

Submitted By: -

SAGAR SUTAR (19CHE014)

Chemical Engineering

GIET UNIVERSITY,GUNUPUR

TRAINING REPORT ON
VOCATIONAL TRAINING
Duration:15th May 22 to 30th June 22
DAP MANUFACTURING PLANT
IFFCO PARADEEP



**GANDHI INSTITUTE OF ENGINEERING
& TECHNOLOGY UNIVERSITY GUNUPUR**

Submitted to: -

K.P. NANDA (Sec. Head Training)

IFFCO Pradeep

Jagatsinghpur, ODISHA

Submitted By: -

SAGAR SUTAR (19CHE014)

Chemical Engineering

GIET UNIVERSITY,GUNUPUR

ACKNOWLEDGEMENT

Industrial Training is an integral part of engineering curriculum providing engineers with first hand and practical aspects of their studies. It gives them the knowledge about the work and circumstances existing in the company. The preparation of this report would not have been possible without the valuable contribution of the IFFCO family comprising of several experienced engineers in their respective field of work. It gives me great pleasure in completing my training at DAP Production Plant of IFFCO at Paradip and submitting the training report for the same.

I express my deepest gratitude to **Mr. K.R.N.MURTY (JGM PRODUCTION)** for giving us the permission for orientation in operational area of plant. I am also thankful to **Mr. K.P. NANDA (Sec Head TRAINING)** who supported us constantly and channelize our work toward more positive manner.

I want to thank **Mr.B. Samantray , Mr.R.K Sahu , Mr. B.C.Behera , Mr. Ranjit ku. Sambana** sir, & **Mr. Bharat ch. Nayak** sir who helped a lot to go through the detail process of DAP & PAP.

Last but not the least our **librarian Mr.Barik** sir who provided us every material we needed. Other sources include manual from library and internet sites to know few details.

A major contribution of this work would definitely be my parents who have constantly supported me for my training in here and my friends who have always been there as a pillar of strength.

And at last but not least we are also thankful to all the staff members of plant for their kind cooperation and valuable guidance throughout the process of work.

PREFACE

In any organization success or failure of the company depend upon 4 M's i.e. Materials, Men, Machine and Method. Today is the age of competition and an organization cannot survive without satisfaction of its customers. Quality of material is to be maintained in order to stand in the competitive market.

To be a perfect engineer one must be familiar with individual experience in industrial environment. He must be aware of basic industrial problems and their remedies.

While undergoing this type of industrial training at **IFFCO PARADIP, Jagatsinghpur , (ODISHA)**. I learned a lot of practical aspect. My theoretical knowledge was exposed here practically. In this report I have tried to summarize what I have learned in the IFFCO plant. For preparing this report I visited the plant, referred to the process and cleared related doubts to the responsible personal & inferred to manuals and process reports.

This study has been primarily undertaken by me with a view to evaluate proper working process in the organization

CONTENT

1. INTRODUCTION

2. FIRE AND SAFETY

3. ABOUT IFFCO PARADIP

4. BRIEF INTRODUCTION AT DAP

5. PROCESSES INVOLVED IN DAP

- **RAW MATERIALS**
- **REACTION INVOLVED**
- **PRENEUTRALIZER**
- **AMMONIA VAPORIZATION**
 - **AMMONIA PRE HEATER**
 - **TAIL GAS VAPORIZER**
 - **AMMONIA VAPORIZER**
 - **GRANULATION**
 - **DRYING AND COMBUSTION CHAMBER**
 - **SCREENING AND CRUSHING**
 - **DUST AND FUMES COLLECTION**
 - **SCRUBBING**
 - **PLANT EFFLUENTS**

EQUIPMENT AND SPECIFICATION

6. INTRODUCTION ON PAP

- **Plant Production Capacity & Design**
- **Product Specification**
- **Process Technology & Process Description**
- **Rock phosphate storage & Conveying Section**
- **Grinding**

- **Reactor & Fumes Scrubber**
- **Single tank reactor**
- **Vacuum cooler**
- **Fume scrubber**
- **Belt Filter & Filtration**
- **Concentration & Fluorine recovery section**
- **Acid storage area & clarification**
- **Cooling Tower**
- **Gypsum pond**

7. CONCLUSION

INTRODUCTION

Indian Farmers Fertilizer Cooperative Ltd. (IFFCO) is one of the largest manufacturers of Nitrogenous and Phosphatic Fertilizer in the world, producing around 8.5 million tonnes of fertilizer in a year. IFFCO is a giant

Cooperative owned by the farmers of India and the fertilizer produced is sold to the farmers through the Farmer Cooperatives. IFFCO is in the services of farmers since 1967. The Society has grown in strength and stature from a modest membership of 57 in 1967-68 to around 40,000 societies, benefiting 5 crore farmers. IFFCO has five manufacturing units in the country. It has diversified into many other business areas like Insurance, Power, Kisan Sanchar, Agro based SEZ etc. It also has operations in many industries abroad like OMIFCO, ICS-Senegal, JIFCO etc. The main objective of IFFCO is to provide quality fertilizer at the door-step of farmers and provide services to boost the output of agro products by the farmers. IFFCO has 3 Nitrogenous and 2 Phosphatic Fertilizer complexes.

PLANTS COMMISSIONED BY IFFCO:

- Ammonia - urea complex at Kalol in the state Gujarat in 1975.
- NPK/DAP plant at Kandla in the state of Gujarat in 1975.
- Ammonia - urea complex was at Phulpur in Uttar Pradesh in 1981.
- The ammonia - urea unit at Aonla was commissioned in 1988.
- DAP/NPK Phosphatic fertilizer unit at Paradip in Orissa in 2005

IFFCO has made strategic investments in several joint ventures.

- Indian Potash Ltd (IPL) in India
- Industries Chimiques du Senegal (ICS) in Senegal,
- Oman India Fertiliser Company (OMIFCO) in Oman
- Jordan India Fertilizer Company (JIFCO)

FIRE AND SAFETY

INTRODUCTION:

- Safety is the 1st step in every organization.

- Fire is the most common serious hazard that one faces in a typical chemistry laboratory.
- While proper procedure and training can minimize the chances of an accidental fire, you must still be prepared to deal with a fire emergency.

BASIC TYPES OF FIRE EXTINGUISHERS:

- The two most common types of extinguishers in the laboratories are pressurized dry chemical (Type BC or ABC, left) and carbon dioxide (CO₂, right) extinguishers.
- We may have other "clean agent " extinguishers besides CO₂, particularly if the workplace has sensitive electronic devices such as computers.
- Those who work with flammable metals may also have a specialized Class D dry powder extinguisher for use on fires (in a pinch, a bucket of dry sand will do, but you really should have Class D unit if you work with such materials). Water filled extinguishers are not acceptable for chemistry laboratory use.

CLASSES OF FIRE:

- **Class A** fires are ordinary materials like burning paper, lumber, cardboard, plastics ,etc.
- **Class B** fires involve flammable or combustible liquids such as gasoline, kerosene and common organic solvents used in the laboratory.
- **Class C** fires involve energized electrical equipment, such as appliances, switches, panel boxes, power tools, hot plates, and stirrers. Water can be a dangerous extinguishing medium for class C fires because of the risk of electrical shock unless a specialized water mist extinguisher is used.
- **Class D** fires involve combustible metals such as magnesium, titanium, potassium, and sodium as well as pyrophoric organometallic reagents such as alkylolithiums, Grignard's and diethyl zinc. These materials burn

at high temperatures and will react violently with water, air, and/or other chemicals.

USING A FIRE EXTINGUISHER:

Use a mental checklist to make a Fight-or-Flight Decision. Attempt extinguisher only if ALL of the following apply :

- The building is being evacuated (Fire alarm is pulled)
- The fire department is being called (dial 101)
- The fire is small, contained and not spreading beyond its starting point
- The exit is clear there is no imminent peril and you can fight the fire with your back to the exit
- We can stay low and avoid the smoke
- The proper extinguisher is immediately at hand
- We have read the instruction and know how to use the extinguisher

GENERAL SAFETY RULES:

- Employees must report all accidents immediately to their supervisor.
- Any defective or unsafe equipment must be reported to a supervisor so that it may be repaired promptly.
- Conform to all safety rules, signs and regulations. If the rules are not understood then a supervisor shall explain them.
- Any unsafe conditions must be reported to a supervisor and corrected, if within the person's responsibility.
- Report to a supervisor any unsafe acts witnessed.
- Report to a supervisor any suggestions you may have for improving safety.
- A safety belt with life line properly attached, anchored and adjusted must be worn when working at a location without guardrails or substantial barriers, if that location is more than 2 mtrs above a lower floor or ground level.
- Childish games or pranks must not be engaged in on the job.

- Access to fire extinguishers or other firefighting equipment must not be blocked. Machine guards must be in position before startup and during normal operation of equipment.
- Equipment must not be operated in an unsafe condition.
- Equipment must not be operated if it will cause unsafe condition.
- Only correct tools (in safe condition) for the job shall be used. (defective or damaged tools must not be used).
- If another employee or visitor is seen performing an unsafe act, stop him and tactfully remind him of the safe way.
- Electrical repairing must only be made by authorized personnel.
- Personnel must not operate any equipment unless they have been authorized to do
- Never watch any type of welding without wearing protective equipment.
- Makeshift scaffolds, ladders, ropes, blocks, cables, or defective slings must not be used.
- The proper tools and equipment for the job must always be obtained and used.
- Objects must not be thrown or dropped from high place unless a watchman has been posted below and if possible, the area has been roped or blocked off with appropriate signs.
- Compressed air must not be used to blow off any part of the body or any clothing while it is being worn, except at authorized locations.
- Alcoholic beverages and drugs must not be allowed on the job. No one shall be permitted to work while under influence of alcoholic beverages or drugs.
- Gasoline shall be used for fuel in motor vehicles and similar equipment, and may not, under any circumstances, used for cleaning purposes or starting fires.
- All parts of the body must be kept away from pinch points when on or near moving equipment.
- Wrist watches, rings, or other metal jewelry must not be worn while working on or near energized circuits.
- Fighting will not be tolerated.
- Personnel must walk, not run, except in event of extreme emergency. When going up or down the stairways, the handrails provided shall be used.

- Shirts and long trousers or equivalent shall be worn. Loose or torn clothing may be caught in machinery and shall not be permitted.
- Carbon tetrachloride must not be used in plant. Contact with electricity may produce poisonous chlorine gas.
- Appropriate safety gear must always be used when required.
- Vessels, tanks, or pits must not be entered without strict adherence to vessel entry procedures.
- Tag and lock out procedure must be strictly adhered to.
- Access to safety and eye wash fountains must not be blocked or obstructed.
- Hot work permit requirements must be adhered to.
- Only low voltage (12 volts or less) explosion proof extension lights shall be used in tanks, vessels, boilers, etc.
- Safety glasses must be worn at all times in designated areas of the plant complex.
- Safety hard hats must be worn at all times in designated areas of the plant complex.
- Safety shoes must be worn at all times within the plant complex.
- Work areas must be kept clean and orderly

IFFCO PARADIP

The Plant at Paradeep, located in Orissa, has been taken over by IFFCO in 2005, which is producing DAP, NP, NPK Gr.I & Gr II Phosphatic Fertilizer. The Phosphoric Acid Plant at is world's largest Plant. IFFCO operates two 3500 TPD Sulphuric Acid Plant, 2650 TPD Phosphoric Acid Plant, 3 X 2090 DAP/NP / NPK Plant, 2 X 55 MW Power Plant and various offsite facilities.

The Plant was commissioned in 2000 but the same was closed due to environmental issues by the Control Board. The Plant was in bad shape during take over and it was not possible to resume production without major revamping of the plant.

In view of the above, IFFCO made a revamping strategy in 2005 so that some part of the plant could be made operational at the earliest. The problem of revamping included environmental issues, safety aspects and reliability of the plant. IFFCO engaged their expert professionals from various plants along with leading Consultants, to identify the problems and its remedies so that the plant, after revamping, can be made similar to the best operating plants,

IFFCO laid emphasis to improve the infrastructure in the plant to ensure that the plants run smoothly without any safety hazard. Today, IFFCO is exporting its surplus power to GRIDCO, Orissa after meeting its own requirement. Phospho-Gypsum is sold to cement plants and for agricultural purposes in various parts of the country. IFFCO has also started sale of Hydrofluosilicic Acid which is a pollutant, to nearby industries, to recover fluorine. Tree plantation and horticulture activities have been given prime importance with a view to make a Green IFFCO.

Welfare of the employees is also given due importance by the management. All the regular employees numbering 1330 (As on 31/03/2013). After taking over, IFFCO has developed a full-fledged township with latest facilities including a well equipped Hospital with 3 Doctors, Recreation Club, School upto 12" Standard, etc.

Under the Corporate Social Responsibility also, IFFCO is carrying out several social activities in the areas of Community development, environment protection, health care/ medical facilities etc. Under the CSR programme IFFCO

is rendering assistance to various Schools, Hospitals, construction of roads, school buildings, tree plantation etc.

After takeover IFFCO has gone for integrated management system consisting of both ISO 14001: 2004 & OHSAS 18001:2007 certification.

Thus, IFFCO has made a total Turn-around of the sick fertilizer complex into a profit making Unit, thereby showing a success-story of acquisition of industry from private sector to the cooperative sector. This level of achievement could be possible only due to dedicated efforts, commitment and team work of the employees. We are confident that with its strong cooperative character, IFFCO will continue to lead from the front with its pioneering efforts and strong financial base.

The unit was originally constructed by M/s. Oswal Chemicals and Fertilizers Limited (OCFL) and commissioned in the year 2000. The plant was operated under OCFL management till take over by IFFCO in October, 2005

Plant capacities, process suppliers and raw materials requirement

PARTICULARS	DAP 3 Streams	PAP Single stream	SAP 2 Streams
Technology			
Process suppliers	M/s Jacobs Engg., Group Inc. USA	M/s Jacobs Engg., Group Inc. USA	M/s Lurgi GmbH Germany

Installation capacities

Particulars	Name of product	Daily Capacities (MTPD)(100% H2SO4)		On Stream Days		Annual Capacity, MT (100% H2SO4)		
		Trains		Trains		Trains		total
		Train -1	Train -2	Train -1	Train -2	Train -1	Train -2	
1	2	3		4		5		6
Design Condition	Sulphuric acid	3500	3500	330	330	1155000	1155000	2310000
	Phosphoric acid (100% P2O5)	2650	-	330	-	874500	-	874500

Particulars	Name of DAP/NPK fertilisers	Grade wise Daily Capacity for Each train (MTPD)	On Steam Days for Each Train	Annual capacity (MT of P2O5)
				Train A+B+C
1	2	3	4	5
Design Condition			304.35	802800
	DAP 18:46:00	2090		
	NP 20:20:00	1760		
	NPK 10:26:26	2310		
	NPK 12:32:16	2310		

The annual requirements of the raw materials at rated capacity of the complex are as follows

S.N.	Raw Materials	Annual Requirement	Source
1	Rock phosphate	2,700,000MT	Jordan , Peru , Morocco , Vietnam , Egypt 9i
2	Sulphur	780,000MT	Middle East
3	Ammonia	400,000MT	Middle East
4	Sulphuric Acid	As per the need	Japan , Korea , Australia , Germany , Indegenous
5	MOP	112,000MT	Canada , Peru



BRIEF INTRODUCTION AT DAP

As far as Indian farmer is concerned, IFFCO's NPK/DAP is not just a source of crucial nutrients N, P, K for the crops, but is an integral part of his/her quest for nurturing mother earth. The bountiful crop that results from this care is an enough reason for the graceful bags of IFFCO NPK/DAP bags to be an integral part of the farmers' family.

The two grades of NPK produced by IFFCO, 10:26:26 and 12:32:16, indicating the content of N, P, K proportion, are tailor made to supply the exact composition required for replenishment of the soil. The Indian farmer's confidence and trust stems from the fact that IFFCO's NPK/DAP are merely a part of a complete package of services, ably supported by a dedicated team of qualified personnel. More importantly, they are aware, IFFCO is a cooperative society owned by farmers cooperatives

Process : Jacob's Engineering Inc (USA)
Process license : Jacob's Engineering Inc (USA)
Plant capacity : 3 steams each of DAP
1.92 Million mtones/Annum of DAP + NPK

Features of Technology

- High efficient use of ammonia & Phosphoric acid
- Utilization of heat of reaction to vaporize Ammonia

Diammonium Phosphate

It is the most popular phosphatic fertiliser because of its high analysis and good physical properties. The composition of pure salt of DAP is N-21.19% and P2O5 -53.76%. Fertiliser grade DAP is 18:46:0.

Typical composition of DAP/NPK complex grades (by weight)	DAP
1. Moisture	0.9
2. Diammonium phosphate	87.53
3. Muriate	--
4. Urea	3.09
5. Filler (silica sand)	8.48

Secondary / micro nutrients	DAP
1. Sulphur	0.48
2. Iron as Fe	0.31
3. Aluminum as Al	0.32
4. Calcium as ca	0.12
5. Magnesium as Mg	0.26
6. Zinc as Zn (ppm)	103
7. Copper as Cu	55

PROCESSES INVOLVED IN DAP

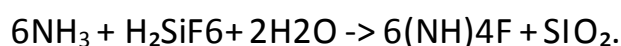
● **Raw materials Required**

- Liquid Ammonia (-33 Deg C) from storage preheated by steam (0-5 Deg C) then sent to the Ammonia vaporizer for attening Temp. of 33-35 Deg C
- Solid raw materials potash , Filler & Urea are transferred to the Fines conveyer by raw material transfer conveyer
- Phosphoric acids at 28% (26-27%) and 54% (46-49%) P2O5 strengths are received from the phosphoric acid plant into their respective storage tanks via wash box.
- From the tanks ,acids are pumped by the 28% P2O5 feed pump and 54% P2O5 feed pump to the various destination of granulation plant. The 54% P2O5 acid is supplied only to the preneutralizer while the 28% P2O5 acid is supplied to the scrubber system and the preneutralizer.

● **Reaction involved**

1. $\text{NH}_3 + \text{H}_2\text{PO}_4 \rightarrow (\text{NH})\text{H}_2\text{PO}_4 + \text{Heat}$
2. $\text{NH}_3 + (\text{NH}_4)\text{H}_2\text{PO}_4 \rightarrow (\text{NH}_4)_2\text{HPO}_4 + \text{Heat}$
3. $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4 + \text{Heat}$

Wet process Phosphoric Acid also contains Fluosilicic Acid (H_2SiF_6), which reacts with NHS as



The other impurities present in the Phosphoric Acid like Al_2O_3 , Fe_2O_3 , MgO and CaO also react with ammonia and Phosphoric Acid to form different compounds, which contribute towards N and P_2O_5 in the product.

Side Reaction

- $\text{H}_2\text{SiF}_6 + 2\text{NH}_3 = (\text{NH}_4)\text{SiF}_6$
- $(\text{NH}_4)\text{SiF}_6 + 4\text{NH}_3 + 2\text{H}_2\text{O} = 6\text{NH}_4\text{F} + \text{SiO}_2$
- $\text{CaO} + \text{H}_2\text{SO}_4 + 2\text{H}_2\text{O} = \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- Other products are $\text{NH}_4(\text{FeH})(\text{HPO}_4)_2$, $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$, MgNH_4PO_4

● Process

Phosphoric Acid and Ammonia react in the Pre-Neutraliser along with scrubber liquor from the scrubbing system. The feed rates of the acid, NH_3 and Scrubber Liquor are maintained such that 1.4 to 1.45 Mole Ratio is achieved in the Pre-Neutraliser. The heat evolved due to the exothermic nature of the reaction is utilized to evaporate considerable quantity of water from Pre-Neutraliser. These fumes along with some ammonia slip pass through the scrubbing system. The reaction slurry of Ammonium Phosphate is pumped vide "Slurry Pump" to Pipe Reactor (PCR) situated in granulator. Further, ammonia is added into Pipe Reactor to ensure the slurry at Mole Ratio 1.80.

→ THE PRENEUTRALIZER

Pre neutralizer is basically a CSTR (Continuous Stirred Tank Reactor) whose sole purpose is to mix the raw materials. The reaction between ammonia vapor and phosphoric acid begins in the Pre-neutralizer. Jacobs uses the reduced retention time in Pre-neutralizer where the diameter at the bottom of the tank is smaller than the top. Pre-neutralizer mix Phosphoric Acid, Ammonia and Scrubber Liquor to be sent to the pipe reactor or directly to the granulator.



The preneutralizer process variables to be controlled are:

- Acid(phosphoric and sulphuric) feed rates , Ammonia feed rate , Slurry N:P mole ratio , Slurry density , Slurry temp & Preneutralizer level.

There are three sources of phosphoric acid to the pre neutralizer:

- 54% (46-49%) P_2O_5 acid which is pumped from storage.
- 28% (26-27%) P_2O_5 acid which is pumped from storage.
- Scrubber liquor[containing recovered fertilizer, dust, free water, weak phosphoric acid, and unreacted ammonia from granulator & P.N] recovered from the prescrubber circulating liquor which is pumped to the preneutralizer by the prescrubber pump

Sulphuric acid at 98% concentration is received from SAP unit to preneutralizer. Sulphuric acid is added for grade control when producing 18-46-00(DAP) and 20-20-00(NP).

Vapour Ammonia from ammonia vapourization is sent to preneutralizer controlled by 8 individual sparger pipes below the liquid level. The ammonia flow is controlled to adjust the N:P mole ratio of the slurry.

The slurry density is used as an indication of its concentration and percent P₂O₅ in the preneutralizer.

The slurry N:P mole ratio is determined from samples taken every hour. The control of the mole ratio is critical in maintaining a pumpable slurry to the granulator, avoiding regions where solidification of slurry can occur and avoiding high ammonia losses.

The preneutralizer slurry temperature is about 118-120 °C when producing DAP and NP is 115.5 °C to 116.5 °C. The slurry is considered to be at its boiling point. The temperature of the slurry is increased with increase in concentration of ammonium phosphates and vice versa.

→ **AMMONIA VAPORIZATION**

There are 3 heat exchangers in which ammonia vaporization can occur. In addition there is a preheater in which liquid ammonia is warmed from -33 °C to 5 °C without vaporization.

Ammonia vaporization will occur in the tail gas vaporizer, which utilizes residual heat from tail gas scrubber liquor

The third vaporizer is the steam heated ammonia vaporizer. When the tail gas vaporizer is operating, there will be little or no load on this unit. However it has the capacity to handle the entire vaporization load if the tail gas vaporizer is out of service for any reason.

• **AMMONIA PREHEATER**

All ammonia fed to the plant will flow first through the preheater. Ammonia flow will be controlled by the flow and level controller of the vaporization units downstream and by the process demand for the grade and the rate being produced.

Low pressure steam flow to the preheater is controlled by the temperature of the condensate leaving the preheater. If the ammonia temperature leaving the preheater is too low, the ammonia controlled valves feeding each of the 3 vaporizer will close.

- **TAIL GAS VAPORIZER:**

It has the dual advantage of greatly reducing steam requirements while also reducing fluoride emissions from the stack. Two of the tail gas scrubber pumps will normally be in operation circulating scrubber liquor from the base of the scrubber through the vaporizer and back to the spray nozzles in the gas inlet duct to the scrubber. A 3 way valve controls the scrubber liquor flow to the vaporizer based on a signal from a pressure controller sensing ammonia pressure in the vapor discharge. This is a split range controller and when the selector switch is set to the normal position, steam will begin flowing to the ammonia vaporizer when the 3 way valve is directing all of the scrubber liquor flow to the tail gas vaporizer. In this way, the ammonia vaporizer supplies supplemental heat as required for vaporization when the tail gas vaporizer is at full capacity.

Liquid ammonia flow to the vaporizer is controlled by the liquid level in the vaporizer. If the level rises too high, the ammonia inlet valves to the TG vaporizer will close.



- **AMMONIA VAPORIZER:**

Normally the vaporization requirements in this unit are small since it serves to supplement the tail gas vaporizer as needed. In this mode of operation, the steam flow is controlled by the smaller of the 2 condensate control valves. In the event that the TG vaporizer is out of service, the mode selector switch is set to the position corresponding to use of the ammonia vaporizer only.



→ **GRANULATION**

The granulation system's job is to turn the slurry into granular fertiliser products with the right content and size. Phosphate slurry is sprayed over a bed of recycled dry material returned by the fines conveyer during granulation in the drum granulator. The granulator's rolling

motion properly distributes the slurry over the granules' surface, and recirculation generates highly homogeneous, hard, well-rounded layered granules.

Agglomerates or thick slurry are more difficult to dry than the thin sheet of slurry that results.

The slurry spray header in the granulator is loaded above the ammonia sparger support pipe so that it does not build up during operation. The granulator's internal wall is lined with flexible rubber panels to prevent accumulation and the creation of big lumps. A scraper driven by an air cylinder is also included to avoid excessive accumulation on the ammonia sparger support pipe. Slurry spray nozzles are also designed in a unique way to avoid sealing and blocking. A number of massive support arms hold the ammonia sparger.

The pre-neutralizer's ammonia reaction is finished in the granulator, resulting in the final product mole ratio.

A single sparger pipe positioned beneath the tumbling bed of particles feeds ammonia vapour from the ammonia vaporisation system to the granulator.



FERTILISER	MOLE RATIO	MOISTURE, WT%
DAP pre-neutralizer	1.45-1.48	20-25
DAP-granulator	1.8-1.85	3.5
20-20-0- pre-neutralizer	1.1-1.5	20-22
20-20-0-granulator	1.6-1.65	2.5-3
10-26-26- pre-neutralizer	1.38-1.4	20-25

Over ammonisation results in a dry discharge stream and increases ammonia loss from the granulator, which raises the scrubber system's mole ratio. Increased development of overgrowth and wetting of the granulation bed suggest a lack of ammonia.

The constant speed fines capture all of the dry recycling solids, which are then returned to the granulator through the secondary elevator.

→ DRYING

The fertiliser is raised and cascaded from the combustion chamber into the dryer by a co-current flow of air quenched combustion gases. The dryer is made to be as efficient as possible while minimising material build-up.

As the hot gas melts the particles from the granulator, the dryer contains a quench air fan to offer positive control of the temperature in the combustion chamber, which is controlled as needed by product grade to avoid melting, ammonia evolution, and fume production.

The actual regulation of the firing rate is based on the temperature of the output gas. Normally, the maximum solids temperature in a DAP dryer is around 80-85 degrees Celsius.

The solids that exit the dryer fall to the dryer discharge conveyor, which discharges to the primary elevator via gravity. Tramp metal that may clog and damage the elevator diverters or screens is removed by an electromagnetic separator hung above the dryer discharge conveyor.

A co-current flow of hot air in the rotary dryer dries the particles discharged from the granulator to a moisture content of 1.0-1.5 wt percent.

The dryer outlet gas temperature controller controls the flow of furnace oil to the combustion chamber burners. A ratio controller controls the combustion air to furnace ratio, which controls the combustion airflow and the combustion air fan discharge damper. The quench air fan discharge damper regulates the flow of quench air to the burner to maintain the required input air temperature.

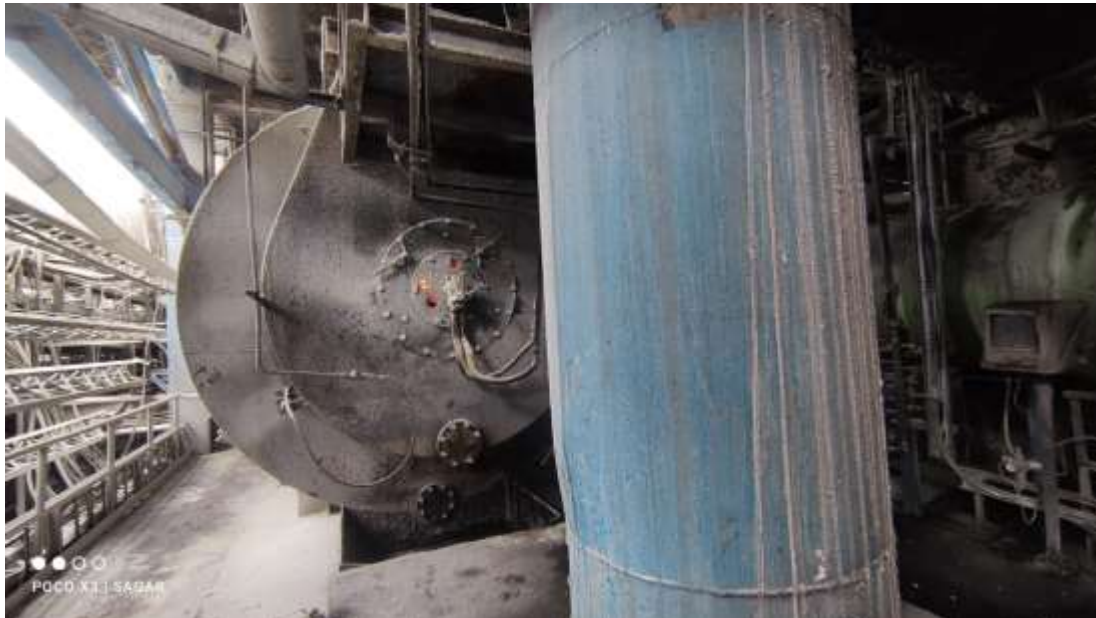
It is required to maintain a modest negative pressure of -2mm WG at the dryer outlet to prevent hot gases from leaking into the building.



COMBUSTION CHAMBER

It comprises of a gun into which furnace oil is injected. Instrument air is used to push the oil, which transports it to the gun's mouth, where it is atomized. There is a spark ignition, which causes the oil to burn. However, because oxygen is necessary for burning, combustion air is introduced. The temperature rises to roughly 1600-1700 degrees Celsius during combustion. To lower the temperature, quench air is supplied, which cools the air to 300 degrees Celsius. The moisture content of the product is reduced from 2.5 percent to 1.5 percent by passing it through

a drier. The product is moved from the dryer to the DDC [dryer discharge conveyor]. There is a magnet in DDC that removes any iron components from the product. It then goes to the major elevator, which transports the goods to the oversize screening area.



→ SCREENING AND CRUSHING

Four single deck motors vibrated large screens are used in each granulation train. All recyclables are filtered by these filters.

A series of chutes and diverters feeds the bigger screens from the principal elevator. A part of the fines fraction from each oversize screen is metered to the four product screens through a product surge hopper variable speed conveyor. Only the amount of -4mm material necessary to deliver the right quantity of product is screened in this configuration. This approach is based on the idea that a major portion of the recycled material is product size. It is desirable for the granulator bed to be fluid in order to achieve maximum production.

Each of the four oversize pulverizes crushes the oversize from each enormous screen. The fines conveyor, which is a large, slow-speed bed with a specifically built cage to prevent dusting, receives the discharge from the chain mill.

To manage granulation and the amount of recycling in the system, the product take-off rate is controlled. This is done automatically by instrumentation reaction to load on the fines conveyor's recycling scale. The cycle's output is utilized to adjust the product screen feeder's speed, which can be broad, slow, or variable.



The load on the product screen is decreased since only the required quantity of particles is screened further, enhancing efficiency and lowering maintenance.

The product screen elevator transports solids from the product screen feeder to the product screens, which are four single-deck vibrated screens that operate in simultaneously.

If the product surge hopper overflows, the overflow material will be sent to the fines conveyor. After being separated from the product on the product screen, the undersize material is sent to the fines conveyor. Through the product elevator, the leftover material is transported to the bagging conveyor. The material from the fines conveyor is sent to the granulator by a secondary elevator. The granulator is then sent to the dryer. The recycling process is so continued.

→ DUST AND FUME COLLECTION

To expel unreacted ammonia, other gases, and dust from the process, all process equipment in the plant runs under a slight negative pressure.

The preneutralizer and granulators exhaust air containing ammonia, water vapour, and dust to a pre-scrubber, where a large percentage of the ammonia is removed by interacting with phosphoric acid in a circulating scrubber solution. The gas passes from the pre-scrubber to the dust and fume scrubber, which removes any remaining ammonia and dust.

Dusty air from solids handling equipment (conveyors, elevators, screens, etc.) is vented into the plant dust collecting system, which then flows to the dust cyclone system, where the majority of the dust is collected and recycled back into the process through the fines conveyor. The air from the cyclone passes to the dust fumes scrubber, where the revolving scrubber solution removes any remaining dust.

The dryer exhaust fan and the dust and fumes exhaust fan, respectively, expel air from the dryer scrubber and the dust and fumes scrubber to the tail gas scrubber.

→ SCRUBBING

Four types of scrubbing are used:-

- Pre-scrubber
- Dust and fume scrubber
- Dryer scrubber
- Tail gas scrubber

Through the scrubbing system, gas moves in a series-parallel fashion. The gases from the preneutralizer and granulator pass via the pre-scrubber, the dust and fume scrubber, and finally the tail gas scrubber. In parallel to the tail gas scrubber, drier gases from the dryer cyclone and dust cyclone gases go to the dryer and dust and fume scrubbers, respectively.

The scrubber liquor from the TG scrubber is pumped back to the dryer and dust and fume scrubber (primary scrubber tank), then to the pre-scrubber tank, and finally to the preneutralizer.

Weak phosphoric acid (28 percent P_2O_5) is pumped from storage to the pre-scrubber tank through flow control. Scrubber liquor is purged from the pre-scrubber tank to the preneutralizer through flow control. Scrubber liquor purge from the primary scrubber tank is pumped to the pre-scrubber tank through the flow control. To modify the flow, the pre-scrubber tank level controller sends a signal to the primary scrubber tank flow control. Scrubber liquor with a N:P mole ratio of around 1.3 is fed to the pre-scrubber, which removes a significant quantity of ammonia from the gases.

Similarly, flow control is used to pump weak phosphoric acid (28 percent P_2O_5) from storage to the primary scrubber tank. The primary scrubber tank circulates scrubber fluid with a N:P mole ratio of around 0.6.

Dust and fume scrubbers, as well as a drier The flow to each scrubber is manually controlled. Scrubber liquor purge is pumped from the TG scrubber to the primary scrubber tank in addition to the weak acid.

The TG scrubber receives scrubber liquid from the TG scrubber base. The amount of process water makeup and the alternative flow of purge effluent to the PAP are controlled by the TG scrubber level controller. This flow is normally non-existent.

On flow control, sump liquor, which consists of spills and collected water, is returned to the primary scrubber tank if required. It can also be pumped to the wash liquor.



→ PLANT EFFLUENTS: -

A sump pump collects liquid drains, spills, and washdowns in a plant sump and transports them to the scrubbing system. There is also a wash liquor tank for temporary storage of recoverable wash liquors.

Solid spillage within the plant, as well as temporary storage of in-process solids when equipment must be emptied, are accumulated in a dump area segregated byproduct grade. When this material is reclaimed by front and loader, it is transferred to a scrap hopper and fed to a scrap reclaim conveyor to the RM transfer conveyor, which then discharges to the fines conveyor for return to the process.

The defoamer metering pumps feed defoamer agent as needed from the defoamer tank to the pre-neutralizer, primary scrubber tank, and pre-scrubber to reduce foam.

For the manufacturing of phosphoric acid, Peruvian, Jordanian, and Moroccan rocks are employed. Due to the presence of greater contaminants in Peru rock, more defoamer is employed.

→ EQUIPMENT SPECIFICATION :-

❖ PUMPS

- **WASH LIQUORE PUMPS**

Duty -50m³ per hr , 12.9 m , 1.50 spg
Type – horizontal centrifugal

- **DEAFOMER METERING PUMPS**

Duty – 1=100l/hr , 11.5m , 0.9spg
Type – positive displacement

- **SLURRY PUMPS**

Duty – 120m³ per hr , 52.6m , 1.57 spg
Type – horizontal centrifugal

- **SUMP PUMP**

Duty- 50m³ per hr , 18.8m , 1.5spg
Type – vertical sump pump

- **PRE SCUBBER PUMP**

Duty – 450m³ per hr , 37.4m , 1.31 spg
Type- horizontal centrifugal

- **PRIMARY SCRUBBER PUMP**

Duty – 500m³ per hr , 30.3m , 1.30 spg
Type – horizontal centrifugal

- **28% P2O5 FEED PUMP**

Duty – 65m³ per hr , 22m ,1.31 spg
Type - horizontal centrifugal

- **54%P2O5 FEED PUMP**

Duty – 60m³ per hr , 21.7m , 1.72spg
Type - horizontal centrifugal

❖ FANS & BLOWERS

- **COMBUTION AIR FAN**

Duty – 40000m³ per hr , 31 c , 300mmw.g.
Type – centrifugal type blower

- **QUENCH AIR FAN**

Duty – 150000m³ per hr , 31 c, 100mmw.g.

Type - centrifugal type blower

- **DUST & FUME EXHAUST FAN**

Duty- 260000m³ per hr , 81 c , 800mm

Type – high efficiency , centrifugal

- **DRYER EXHAUST FAN**

Duty – 175000m³ per hr , 58 c , 800mm

Type – high efficiency , centrifugal

❖ **AGITATORS & MIXERS**

- **PRENUTRALIZER AGITATOR**

- **SUMP AGITATOR**

- **GRANULATOR**

Duty- 4,5m ID 9 m long, 8 R.P.m, slope-1:16

Type- rotary drum

- **PRESCRUBBER TANK AGITATOR**

Duty-20 m³ tank, 1.5 spg

Type top entering, 4-45 pitched blade

- **LIQUOR TANK AGITATOR:**

Duty-400 m³ tank, 1.3 spg

Type-top entering, 3 bladed hydrofoil.

- **28% P205 STORAGE TANK AGITATOR:**

Duty- 1400m³ tank, 1.3 spg

Type-top entering, 3 bladed hydrofoil

- **54%P205 STORAGE TANK AGITATOR:**

Duty-1400 m³ tank 1.7 spg

Type-top entering, 3 bladed hydrofoil

❖ **FIRE HEATERS**

- **COMBUSTION CHAMBER**

Duty-13.5 MKcal/hr, fuel oil

Type-direct fire

❖ **TANKS , ATMOSPHERERIC VESSELS**

- **DEFOAMER TANK**

Size-3m dia 3m h, 21 m³

Type-vertical cylindrical

- **PRIMARY SCRUBBER TANK:**

Size-3m dia 4m ht, 35 m³

Type-open top

- **PRESCRUBBER TANK:**

Size-3m dia 3m ht, 20 m³

Type-open top

- **WASH LIQUOR TANK:**

Size-9 m dia 7 m ht, 400 m³

Type-open top

- **28% P205 STORAGE TANK:**

Size-14m dia 10.5m ht, 1400 m³.

Type-closed top

- **54% P205 STORAGE TANK:**

Size-14m dia 10.5m ht, 1400 m³.

Type-closed top

- **PHOSPHORIC ACID WASH BOX:**

Size-2m 4m*1m

Type:- 3 compartment distribution and wash box.

❖ **WET SCRUBBER**

- **PRE-SCRUBBER:**

Size-166000 m³/hr, 91 deg C (max), 100 mm w.g., 4.0m dia

Type-Dust cyclone

- **DRYER SCRUBBER:**

Size 175000 m³/hr. 88 deg C, 300 mm w.g., 4.2m dia.

Type-Venturi cyclonic

- **DUST AND FUME SCRUBBER:**

Size- 260000m³/hr, 76 deg C(max), 300 mm w.g., 5.0 dia

Type-Venturi cyclonic

- **TAIL GAS SCRUBBER:**

Size-562000 m³/hr, 65 deg C(max), 125mm w.g., 7,2m dia
Type-cyclonic

- ❖ **HEAT EXCHANGERS**

- **AMMONIA VAPORIZER:**

Duty-10647000 Kcal/hr, 32tph NH₃
Type- Vertical shell and tube

- **TAIL GAS VAPORIZER:**

Duty-9870000 Kcal/hr, 31.8 tph NH₃
Type-Shell and tube kettle

- **AMMONIA PREHEATER:**

Duty-1221000 Kcal/hr, 32 tph NH₃
Type-shell and tube

- ❖ **REACTORS**

- **PRENEUTRALIZER:**

Size-41 m³, 122 deg C
Top section-6m dia 3.4m ht Bottom section-4.4m dia*
2.5m ht

- ❖ **DRYER**

- Size-680 tph, 4.5m dia 33.7m+1.2m length, 2.5 or 4 rpm slope=1:16
Type- rotary co-current hot gas

- ❖ **OVERSIZE PULVISERS**

Duty- 75tph each
Type-double opposed chain mills

- ❖ **SCREENING**

- **OVERSIZE SCREEN:**

Size-9.3m³, 680 tph(total), 4mm sep
Type-single dec, motor vibrated 36 deg

- **PRODUCT SCREEN:**

Size-9.3m³, 600tph(total), 1mm sep
Type-single dec, motor vibrated 36 deg

- ❖ **BELT CONVEYERS**

- **FINES CONVEYORS:**

Duty- 2m width 40m centers, 680 tph Type-slow speed,
variable speed enclosed

- **DRYER DISCHARGE CONVEYOR:**

Duty-1.4m width*5.4m centers, 680 tph
Type-Inclined belt conveyor

- ❖ **BUCKET ELEVATORS**

- **PRIMARY ELEVATORS:**

Duty-680tph,36m centers
Type-continuous discharge, heavy duty discharge

- **SECONDARY ELEVATORS:**

Duty-680tph,32m centers
Type- continuous discharge,heavy duty discharge

- **PRODUCT ELEVATORS:**

Duty-185 tph,37 centers
Type- continuous discharge heavy duty

- **PRODUCT SCREEN ELEVATORS:**

Duty-600tph,36 centers
Type- continuous discharge heavy duty

PAP MANUFACTURING PLANT



1. INTRODUCTION

Manufacturing phosphoric fertilisers depends heavily on the production of phosphoric acid. For their growth, plants typically need supplements like nitrogen, phosphorus, and potassium. Only nitrogen is added to soil by nitrogenous fertilisers like urea. But because this type of soil also needs phosphorous for growth, phosphorous can be added to the soil by using phosphoric fertilisers (N-P-K-S). The letters N stand for nitrogen, P for P_2O_5 , K for potassium, and S for sulphur. Different grades of phosphatic fertiliser, including 20:20:00:13, 18:46:00, and 14:35:14, can be produced. The application of these grades will vary depending on the needs of the soil. Ammonia and phosphoric acid are necessary for the aforementioned fertilizer's production.

The plant was designed by Jacob's employing Di-hydrate process

The main section of the plants are:

- ❖ Reaction Section
- ❖ Filtration Section
- ❖ Concentration & Fluorine Recovery Section

A powerful acid, typically sulfuric acid, reacts with tricalcium phosphate in rock to produce phosphoric acid and gypsum in the numerous forms of phosphoric acid manufacturing procedures. From places like Togo, Jordan, Vietnam, Egypt,

Peru, and Morocco, among others, rock phosphate is imported. Because rocks come from different sources, their behaviours vary because they have distinct chemical compositions. A combination of rocks from Togo and Morocco will be used in the plant at the IFFCO Paradeep site.

2. Plant Production Capacity & Design:

Capacity of IFFCO Phosphoric acid plant: 2650 MTPD (100% P_2O_5) as weak acid

Capacity of Evaporator product: 2250 MTPD (as strong acid)

Annual stream days: 330

The design of the plant was provided by M/s Jacob's Engineering Group, U.S.A and its. subsidiary, M/s Humphrey & Glasgow Consultants Ltd., provided the detailed Engineering and Construction consulting.

3. Product Specification

Component	Strong	Weak
P_2O_5	52.94	27.72
H_2SO_4	3.21	2.02

4. Process Technology

Di-hydrate Process:

This was the conventional process for most of the 20th century. This process has a long track record or reliable operation, but it lacks the energy efficiency and many of the operating advantages of the Hemi hydrate process.

Phosphate rocks must be finely ground before processing. The filter product Phosphoric Acid is typically only 25-28% P_2O_5 , so further concentration is required before making phosphate fertilizers.

In this process, rock slurry at 65% solid is metered in the first zone of the annular section where it is diluted with the streams re circulated from Cooler Feed Tank. The dilution effect is large because the rock flow is much smaller than the re circulating stream. The rock must be dispersed

well to prevent rock coating and so to minimize phosphate substitution losses.

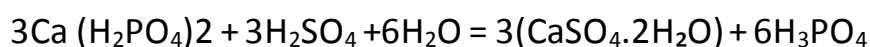
Here 1st reaction takes place and a solution of Mono calcium Phosphate is formed.



Tri calcium Phosphate + Phosphoric Acid Mono calcium Phosphate

Sulphuric Acid is metered through mixing tee with dilute phosphoric acid, coming from the filtration section, to facilitate its dispersion. High local concentrations of Sulphuric acid are undesirable because they cause nucleation and crystal dehydration.

Second reaction takes place between sulphuric acid and mono calcium phosphate to form Phosphoric Acid and Gypsum.



Mono calcium Phosphate + Sulfuric Acid + Water Gypsum Phosphoric Acid

Di-hydrate process produces gypsum in the form of calcium sulphate with of water (H₂O) i.e, CaSO₄.2H₂O or Calcium sulphate di-hydrate). molecules

5. Process Description

Rock phosphate storage & Conveying Section:

Three rock phosphate storage sheds A, B and C of capacities for storing rock phosphate exists. The storage sheds A, B have 75,000 MT each and C has 1, 00,000MT capacity. Each storage shed is built with an underground (tunnel) conveyors 103 A,B &C respectively for conveying rock phosphate from storage shed to underground silos A, B and C present in ball mill area. The capacity of each conveyor is 550MTPH. These conveyors discharge the rock phosphate on conveyor 104 which in

turn discharges the material on conveyor 105. The conveyor 105 discharge can be taken in Silo A and on conveyor 106A. Material from 106A is received by Silo B through 106B conveyor. Finally material from 106B is discharged into Silo C.

Grinding:

Rock phosphate storage & Conveying Section:

- Three rock phosphate storage sheds A, B and C of capacities for storing rock phosphate exists.
- Each storage shed is built with an underground (tunnel) conveyors 103 A,B &C respectively for conveying rock phosphate from storage shed to underground silos A, B and C present in ball mill area.
- These conveyors discharge the rock phosphate on conveyor 104 which in turn discharges the material on conveyor 105.
- The conveyor 105 discharge can be taken in Silo A and on conveyor 106A. Material from 106A is received by Silo B through 106B conveyor.

Grinding:

- The rock phosphate received in the storage sheds is coarser and need to be crushed to required size in the ball mills. • There are four ball mills each of 110MTPH capacity.
- The discharge from ball mill is stored in Slurry storage tank providing buffer capacity for ground rock.
- Rock is transferred to ball mills from the corresponding weigh feeder along with recycled material from the corresponding rock screens.
- A provision for admitting process water as feed to ball mill also exists.
- There is a provision of bypassing the screens and allowing the discharge of mill slurry pumps directly into Slurry sludge tank.
- The screens are usually bypassed during the processing of Jordan and Egypt rock source.



Reactor & Fumes Scrubber:

- Rock phosphate slurry and sulphuric acid react in this section to form Phosphoric acid and calcium sulphate di hydrate, CaSO_4 .
- The reaction is designed to produce a gypsum that can be washed and filtered easily resulting in an expected recovery in the reaction and filtration sections of 96.09(wt) of the P_2O_5 fed to the plant.
- Major components in the section are:
 - a. Single tank reactor (2300 m^3 volume)
 - b. The adjustment and Filter feed tank (200 m^3 vol.,
- The cooler feed tank (770 m^3 vol.) Rock digestion primarily occurs in the single tank reactor with very high slurry recirculation and mixing.
- The other vessels provide additional residence time, feed the vacuum cooler, seal the barometric leg of the vacuum cooler and segregate the slurry pumped to the filtration area.
- Total reaction volume is 5120 m^3 , 2300 m^3 in the single tank reactor, 770 m^3 in each of the cooler feed and 1650 m^3 cooler seal tank, 200 m^3 each in the adjustment and filter feed tank.
- These pumps re circulate a total of 36000 m^3/hr of slurry from the last vacuum cooler feed section to the first section of the single tank reactor.

Single tank reactor:

- The reactor is built as two concrete tanks.
- Underflow weir connect section 117 agitator zone to the centre compartment, the centre compartment to the cooler feed compartment and the cooler feed compartment to the cooler feed tank.
- The differences in level, the circulating pumps and the weirs routes the flow from annulus 111 to 117 agitator zone in the annulus, to the centre compartment, to the cooler feed compartment and to the cooler feed tank.
- Most of the flow 34800 out of total of 3600m³/hr is re-circulated from the cooler feed tank back to section 111 agitator.
- Rock slurry at 65% solids is metered into section 1 where it is diluted with the streams re circulated from the cooler feed tank.
- High local concentrations of sulphuric acid are undesirable because they cause nucleation and crystal dehydration.
- Defoamer is added in reactor to control the foam level as excess foam in reactor will lead to carry over in vacuum cooler circulator.



Cooler feed tank:

The cooler feed tank is a rectangular concrete tank that is connected to the cooler feed compartment of the single tank reactor system by an underflow.

Vacuum cooler circulating pumps feed slurry from this tank to each vacuum cooler. This tank is vented to the fumes scrubber.

Vacuum cooler:

- The reaction in the single tank reactor is exothermic.
- The vacuum coolers remove the heat to control reaction temperature by evaporating a small amount of water from the recirculation slurry under a vacuum of about 11.9" Hg.
- Cooling also de supersaturates the filter feed to reduce scaling of the filter.
- The coolers are mounted directly on the cooler feed tank to minimize the head of the cooler circulating pumps.
- A head of slurry above the inlet nozzle prevents flashing and scaling in the inlet line.
- Normally 100% of the total recirculation flow, 36000m³/hr, is pumped through the coolers with the vacuum cooler circulating, 9000 m³/hr each.
- The cooler is designed to limit slurry cooling to about 2.7°C to minimize nucleation and scaling.
- Each vacuum cooler has two barometric condensers, the vacuum cooler pre condenser and the vacuum cooler condenser and a vacuum cooler ejector.
- The ejector uses medium.
- pressure steam as the motive fluid.
- The Pre condenser removes more than 90% of the fluorine released in the vacuum) cooler by absorption in cooling tower.
- The barometric leg of the pre condensers are.
- sealed in vacuum pre condenser seal tank which is vented to the fume scrubber.
- The V.C pre condenser pumps re-circulate the water from the tank through the spray nozzles in the pre condensers.

- The system recovers part of the heat of reaction by heating the cooling water from 35°C to 65°C. The vacuum pre condenser seal tank supplies water to the filtration area for cake, cloth and belt washing.
- To insure that the water is at the temperature required in the filtration area, about 65°C, low pressure steam is sparged into the seal tank.
- A temperature controller regulates the steam flow, The balance of the vacuum vapors about 80% of the total is condensed with reused cooling tower water in the vacuum cooler condensers, cascade type barometric condensers.
- The barometric leg of the condensers is sealed in a tank, vacuum cooler condenser seal tank, which is vented to the fume scrubber.

Cooler seal tank:

The cooler seal tank is a rectangular concrete tank identical in size to the cooler feed tank. The tank seals the down leg of the vacuum coolers. It overflows through a weir to the annular section of the single tank reactor and to the adjustment compartment of the Adjustment and filter feed tank. Since weir to the reactor is lower than the weir to the adjustment compartment, the slurry flows preferentially to the reactor for recirculation. This tank is vented to the fume scrubber.

Adjustment and Filter feed tank: This tank has two compartments 1) the Adjustment. section and 2) the Filter feed section. The adjustment compartment is equipped with a line to add sulphuric acid on flow control to lower the calcium concentration in the filter feed liquor without affecting the sulphate level in the single tank reactor.

Fume scrubber:

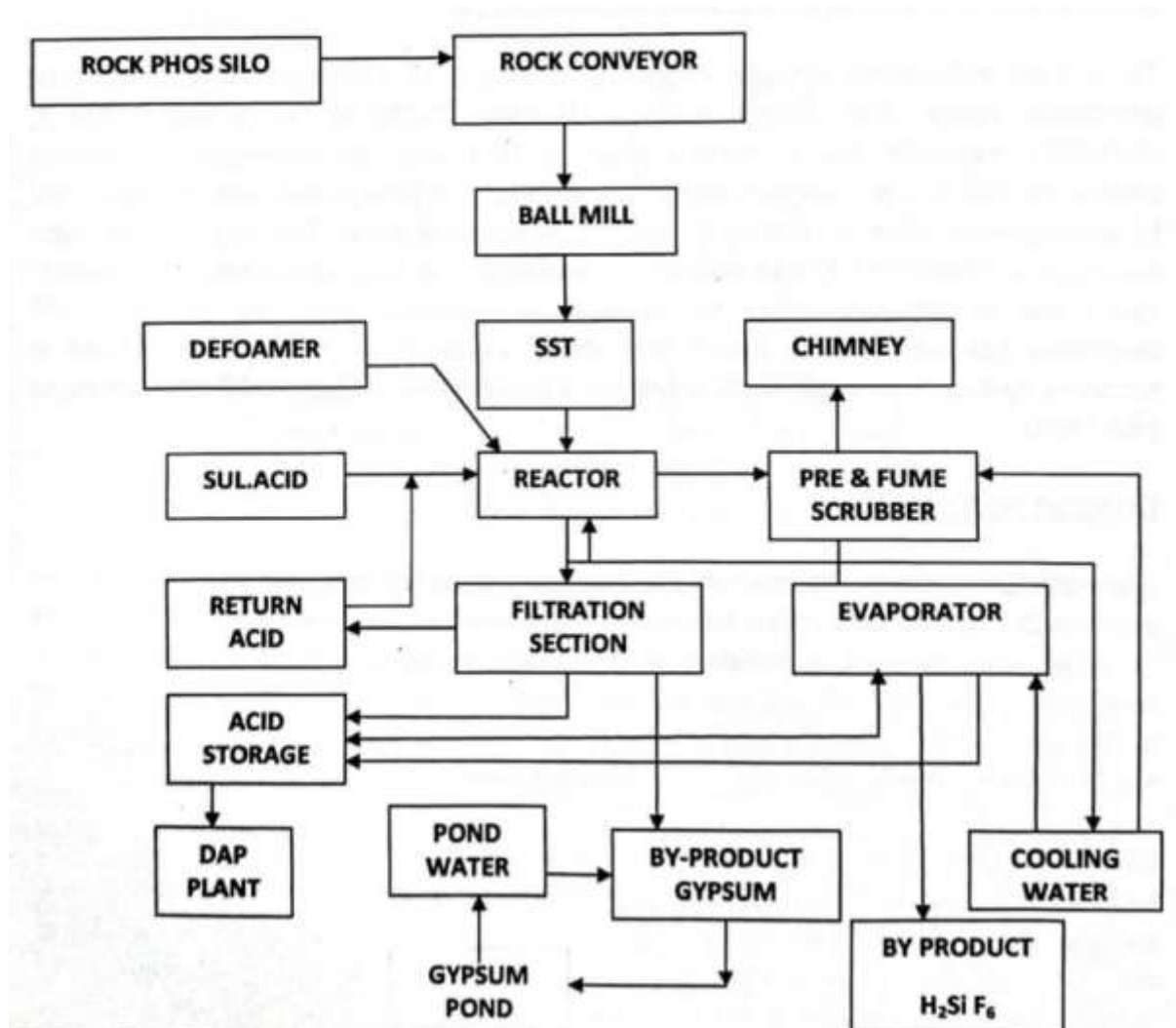
The scrubber reduces the fluoride level in the gas vented through the plant stack to 10g as Fluorine per ton of P_2O_5 , fed to the plant. A plant wide header routes fumes to the Fume scrubber. The scrubber operates at a slight vacuum from about -100 mm H₂O (g) at the inlet to 350mm H₂O at the outlet provided by the scrubber exhaust fan.

Belt Filter & Filtration:

- In the filtration section, seven horizontal belt filters (six in service and one stand by) separate acid and gypsum.
- Filter product containing 26-27% P_2O_5 is pumped to the weak acid clarification section.
- The cloth and the belt return to the front end of the filter through separate but parallel paths.
- The warm water from the vacuum cooler pre condenser seal tank is used as filter wash water.
- The water is received on demand from the spillage sump in filtration area.
- From the cake wash tank it is pumped on to the filter as filter cake wash.
- The first flooded compartment on the belt holds product acid.
- The acid collected from the front end of this compartment is diluted with the cloth wash water retained by the cloth.
- This acid drains through the cloudy filtrate section of the vacuum box to the return acid compartment of the primary filtrate seal tank for recycle to the reactor.
- The balance of the acid from the first flooded compartment drains through the No.1 filtrate section of the vacuum box to the product and compartment of the primary seal tank.
- This acid is the 26-27% P_2O_5 filter product acid which is pumped to the weak acid storage and clarification section.
- The other three flooded compartments on the belt are used for counter current cake.
- washes.
- The fourth flooded compartment normally holds water from the cake wash water pump.
- 4 filtrate compartment of the secondary filtrate tank for use as wash in the third flooded.
- wash compartment.
- The fourth flooded compartment can be washed additionally with water from final wash sump which receives water from recovery sump.

- The seal water outlet of the vacuum pump provides the input source to the recovery sump and water flows from recovery trench to final wash sump.

6. Process Flow Sheet



7. Filtration Section

- The first and second flooded compartments in the filter and the two filtrate seal tanks are vented to the fume scrubber.
- The gas mainly air saturated with water flows to a packed bed condenser, the filter condenser, where most of the water vapor is condensed with process water.
- The barometric leg of the condenser is sealed in the recovered water sump.
- The gas leaving the condenser is compressed with the liquid ring Filter vacuum pump which discharges to the atmosphere.

- Process water/TG blow down water is used as seal water for the pump.
- The gypsum leaving the filter is sluiced in the gypsum cake hopper with pond water or blow down water from fume scrubber sump.
- Gypsum and water flow by gravity in to the gypsum slurry tank where additionally pond water is added to form 25% solid slurry.
- The slurry is pumped to battery limits with the gypsum slurry pumps.

8. Concentration

- In this section, phosphoric acid is concentrated from 27% to 49% P₂O₅ in the evaporator.
- The concentrated acid is pumped to the acid storage and clarification section.
- There are seven evaporators (six running and one stand by).
- Each evaporator is at single stage, forced circulation unit consisting of a rubber lined flash chamber, axial flow evaporator circulation pump and vertical shell and tube, evaporator heat exchanger with.
- Acid is re circulated at a high rate (7,315 m³/hr) by the pump through the heat exchanger and into the flash chamber where an absolute pressure of about 2.6" Hg is maintained to keep the acid at its boiling point.
- The evaporator heat exchanger is a single pass shell and tube unit consisting of 649 tubes vertically mounted in a carbon steel shell with graphite tube sheets.
- At the rated pump capacity, the average acid velocity through the tubes is 2.75 metres per second with an acid temperature rise of 2.6°C.
- The liquid level in the flash chamber is set at a sufficient height to suppress boiling in the tubes which would result in scaling of heat exchanger surface.
- Feed acid enters the evaporator through a nozzle in the chamber at the discharge end of the heat exchanger.
- The lower 1500 mm of the heat exchanger shell is constructed of 316 L SS to provide corrosion resistance in the event of acid contamination of the condensate.
- Overhead vapors from flash chamber pass to an Entrainment separator before fluorine recovery.
-
- Acid removed by entrainment flows back to the suction side of the evaporator circulation pump.

- The acid outlet of the heat exchanger is located below the liquid level in the flash chamber to suppress boiling in the tubes.
- Vacuum is established in the evaporators by admitting cooling tower water into the barometric condenser.
- Earlier vacuum is established by also admitting MP steam in the three stage ejector system connected with barometric condenser.
- When this occurs, a high pressure water pump is used to lance and clean the tubes.

Fluorine recovery section:

The fluorine scrubber is essentially a large void spray tower where the gases leaving the evaporator are passed counter current to a large re circulating flow of fluo-silicic. 12 set of nozzles arranged at three different elevations absorb the HF fumes from the vapor. The tank is vented to the fume scrubber. A partition separates the tank into tow compartments which serve the dual functions of barometric seal and pump tank.

9. Acid storage area & clarification

There are two clarifiers each for weak & strong acid storage

Sl No.	Weak acid clarifier	Vol.	Strong acid clarifier	Vol in m3
1	T-411	2100	T-413	4000
2	T-412	4200	T-414	4200

The tanks are constructed of rubber lined carbon steel with a layer of carbon brick on the bottom. They are equipped with a rubber covered 317 L SS rake, center feed well, bottom sludge discharge cone and clarified acid take off nozzles at three elevations. The rake mechanisms are automatically raised if torque exceeds a pre set value.

Sludge pumps provide the circulation of acid in the tank. Overflow from the weak acid clarifier flows to pump tank of 1100 m³ volume. Weak acid from pump tank will be pumped to evaporators and DAP as per requirement. Similarly

New pump tank serve as pump tank for pumping strong acid to DAP.

The filter acid distribution box is divided into three compartments one compartment towards T-411, other towards T-412 and middle one is wash box zone. Similarly the strong acid feed distribution box is also divided into three compartments one compartment towards T-413, other towards T-414 and middle one is wash box. The feed hoses are diverted to either one of the boxes as per the requirement.

A plate heat exchanger is installed in evaporator section for cooling the strong acid for export to Kandla unit.

10. Cooling Tower

Hot water from the plant flows by gravity to the cooling tower hot well. There are seven hot well pumps and seven cold well pumps. Each pump is rated for about 4,600 m³/hr.

The cooling tower has eight cells each with a stainless steel or glass reinforced vinyl ester fan. The tower has a treated frame with glass reinforced vinyl ester louvers. Filling is PVC on a Glass reinforced Vinyl Ester grid. Cooling water gravitates to the base of the tower which sits above the cold well. There are two pond water lines for adding make up water to cooling tower. All the fan blades were recently replaced with hollow one as part to conserve energy. The duty of the cooling tower is 27,448 m³/hr @ 30°C WB, In/Out 43/35°C, side entry, and top exit with acid brick line concrete basin.

11. Export Tank

The export tank is constructed and situated near Ammonia storage area for exporting strong phosphoric acid. The tank is 40m in diameter and 12 m height with three outlet nozzles. Two pumps are provided for transferring the acid to either the ship or to the strong acid distribution box in PA plant. Pump 444 in PA storage area serves to pump acid from plant to export tank.

Air distributors are provided inside the tank for keeping the solids in suspension. A provision to flush the discharge and suction lines of pumps with air also exists. A sump with a pump to collect the spilled acid back to the export tank also is provided. Pond water line for cleaning and recollection system provision back to PA plant exists.

12. Gypsum pond

There are two ponds for storage of gypsum.

1. East pond
2. 2 West Pond

Slurried gypsum from the plant is pumped to the pond through gypsum pumps in filtration section. The pond consists of three dykes. 1. The inner dyke 2. The outer dyke and 3. The rim dyke. The slurry pumped from the plant flows through the inner dyke towards inside of the stack. Gypsum gets decanted in the middle and water from pond flows out through the rim dyke and enters the outer dyke. The water from the outer dyke enters the surge pond. In the surge pond there are five return water pumps which pump the pond return water to PA plant. As per the plant requirement, pumps will be run. The water level of the surge pond will be monitored on shift basis. In addition to the water from the gypsum stack, surge pond receives water from garland dyke which collects the rain water in the nearby vicinity.

The selection of pumping slurry to the desired pond will depend on the water level over the gypsum pond and stacking activity. Heavy equipment like excavators, cranes, dozer operate to pile up gypsum stacking.

CONCLUSION:

In conclusion. I can honestly say that my time spent during industrial training with IFFCO Paradeep will be valuable experience for me in the future. During 45 days of internship Not only did I gain practical skills but I also had the opportunity to meet many fantastic people. I have been able to meet and make network with so many people that I am sure will be able to help me with opportunities in the future.

One main thing that I have learned through this internship is studying not only in the classroom. This is because the working environment is totally very different from life as a student. Thus, an opportunity for training here has opened my mind to be better prepared for employment after graduation. The other thing that I have learned is time management skills as well as self motivation. Once I realized what I had to do I organized my day and work so that I was not overlapping or wasting my hours. I learned that I needed to be organized and learn how to motivate myself to work under pressure. Last but not list, I learned to respect the others. This attitude is indispensable when working to create a positive working atmosphere.

Overall, my internship at IFFCO Paradeep has been a success. I was able to gain practical skills, work in a fantastic environment, developed my soft skills and make connections that will last a lifetime. I could not be more thankful